

Physical Chemistry (II)

Lecture 11

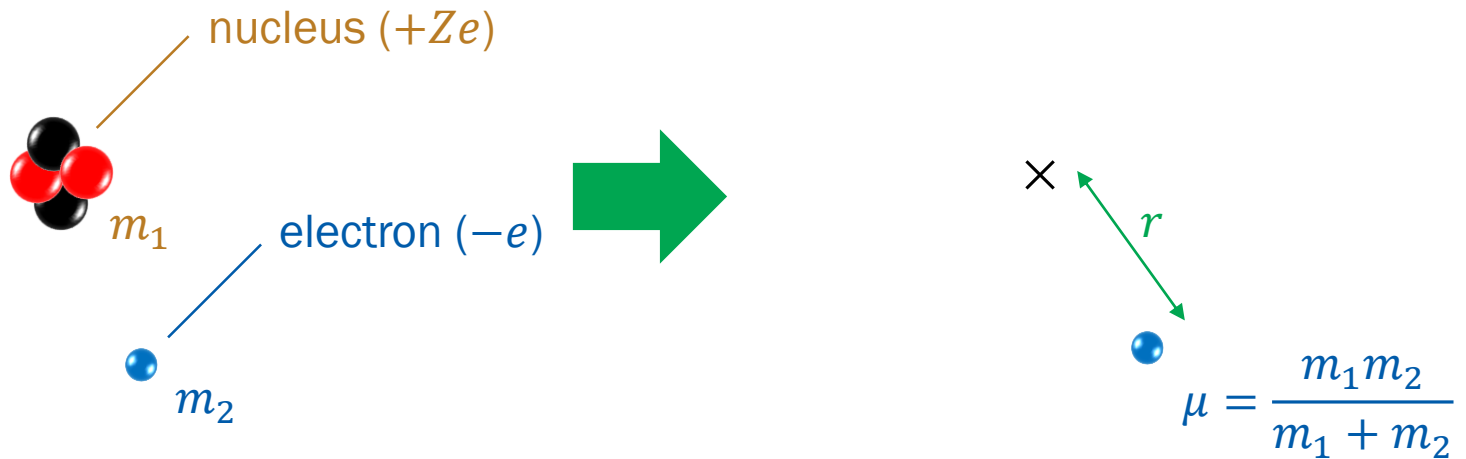
Zeeman Effect



Last class

Hydrogen-like Atom

one nucleus + one electron



$$V(r) = -\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r}$$

Schrödinger Equation

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + \mathcal{V}(\hat{r}) \right] \psi(r, \theta, \phi) = E\psi(r, \theta, \phi)$$

Using separation of variables, $\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$,

$$-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} + \frac{\lambda}{r^2} \right) R(r) + \mathcal{V}(\hat{r})R(r) = ER(r)$$

$$\left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] Y(\theta, \phi) = \lambda Y(\theta, \phi)$$

Solutions

- Quantum numbers: n, l, m_l , such that

$$n = 1, 2, 3, \dots; l = 0, 1, 2, \dots, n - 1; m_l = 0, \pm 1, \pm 2, \dots, \pm l$$

- Wavefunction (a.k.a. “orbital”):

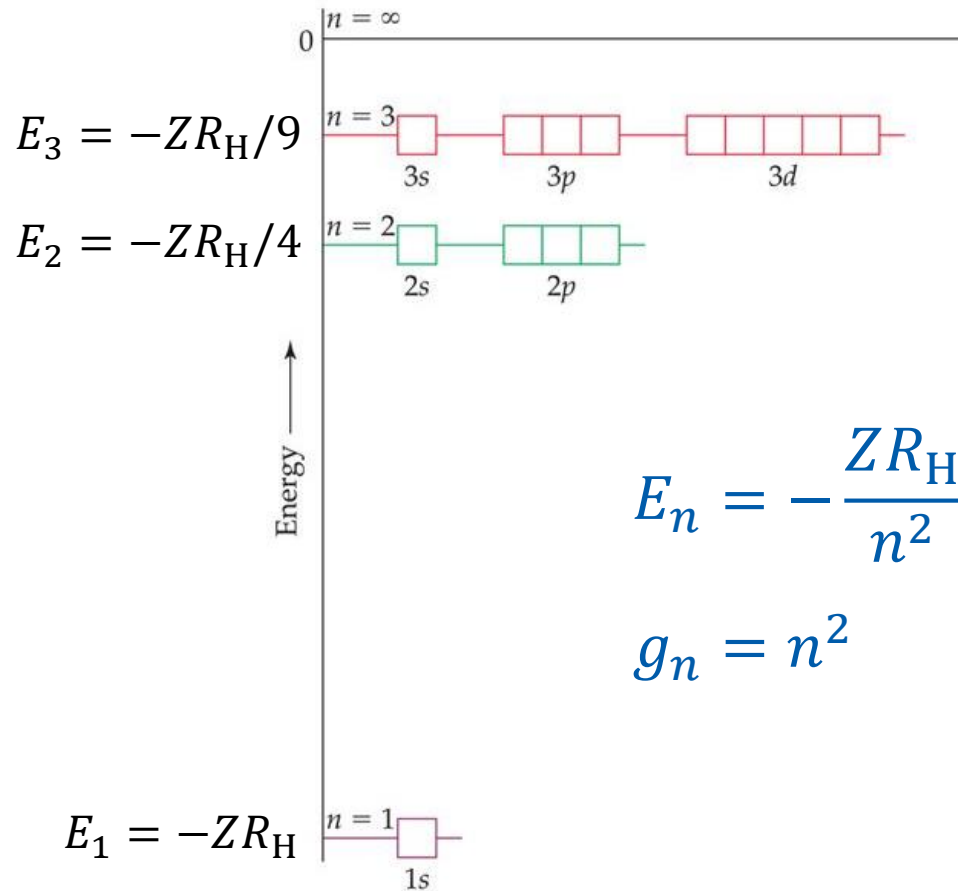
$$\psi_{nlm_l}(r, \theta, \phi) = R_{nl}(r)Y_l^{m_l}(\theta, \phi)$$

- Energy:

$$E_{nlm_l} = -\frac{Ze^2}{8\pi\epsilon_0 a_0} \frac{1}{n^2} = -\frac{ZR_H}{n^2} = E_n$$

Last class

Energy Levels



(Image: <https://slideplayer.com/slide/5959399/>)

Wavefunctions

$$\psi_{nlm_l}(r, \theta, \phi) = R_{nl}(r)Y_l^{m_l}(\theta, \phi)$$

- Normalization condition

$$\int_0^{2\pi} \int_0^{\pi} \int_0^{\infty} \psi^*(r, \theta, \phi) \psi(r, \theta, \phi) r^2 \sin \theta \, dr \, d\theta \, d\phi = 1$$

- Orthogonality condition ($a \neq b$)

$$\int_0^{2\pi} \int_0^{\pi} \int_0^{\infty} \psi_a^*(r, \theta, \phi) \psi_b(r, \theta, \phi) r^2 \sin \theta \, dr \, d\theta \, d\phi = 0$$

Last class

Wavefunctions: Eigenfunctions

$$\psi_{nlm_l}(r, \theta, \phi) = R_{nl}(r)Y_l^{m_l}(\theta, \phi)$$

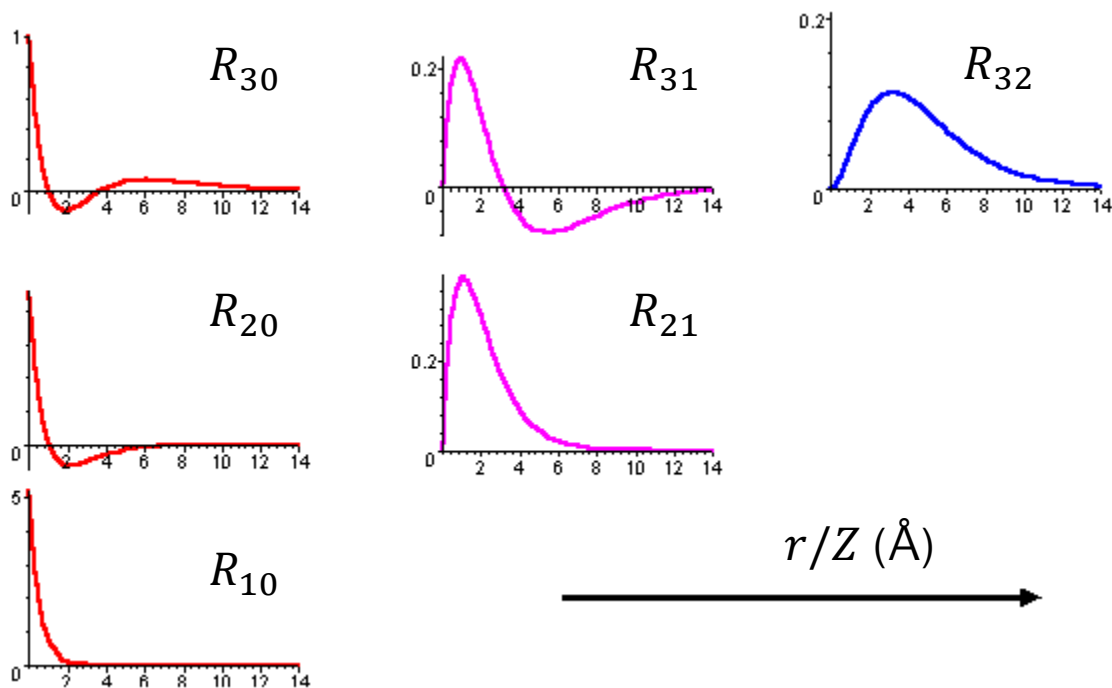
$$\hat{H}\psi_{nlm_l} = E_n\psi_{nlm_l}$$

$$\hat{L}^2\psi_{nlm_l} = l(l+1)\hbar^2\psi_{nlm_l}$$

$$\hat{L}_z\psi_{nlm_l} = m_l\hbar\psi_{nlm_l}$$

Last class

Radial Parts R_{nl}

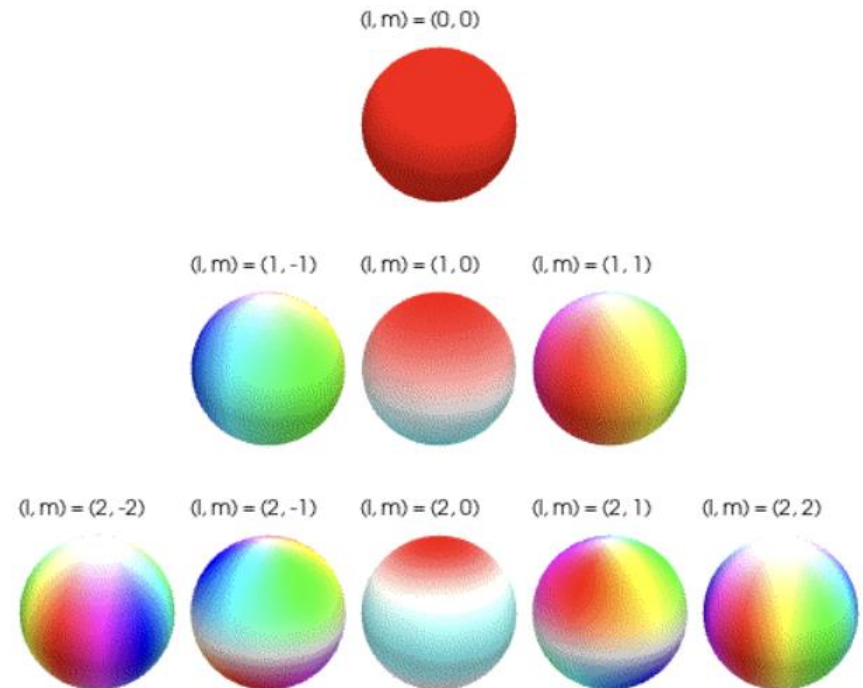
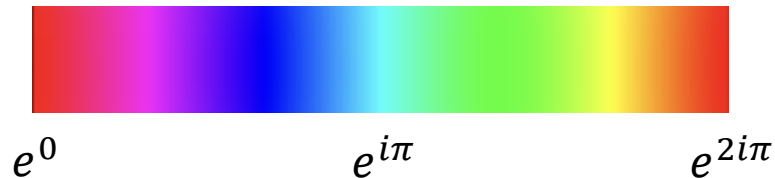


Last class

Angular Parts $Y_l^{m_l}$

$$Y_l^{m_l}(\theta, \phi) = \frac{1}{\sqrt{2\pi}} \Theta_l^{|m_l|}(\theta) e^{im_l\phi}$$

“Hue” indicates the e^{ix} part:

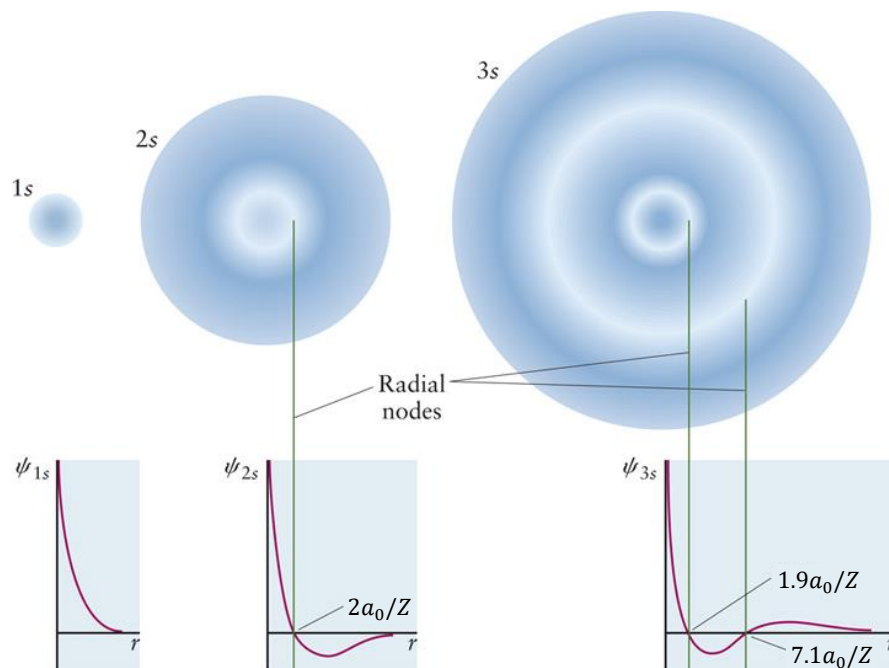


(Image: https://commons.wikimedia.org/wiki/File:Complex_Spherical_Harmonics_Figure_Table_Complex_Polar_Plot.gif)

Last class

s Orbitals

$$\psi_{n00}(r, \theta, \phi) = R_{n0}(r)Y_0^0(\theta, \phi) = \frac{1}{\sqrt{4\pi}} R_{n0}(r)$$



(Image: Oxtoby et al., *Principles of Modern Chemistry*, 7th Ed., Fig. 5.7)

p Orbitals

$$\psi_{n10}(r, \theta, \phi) = R_{n1}(r)Y_1^0(\theta, \phi) = \left(\frac{3}{4\pi}\right)^{1/2} R_{n1}(r) \cos \theta$$

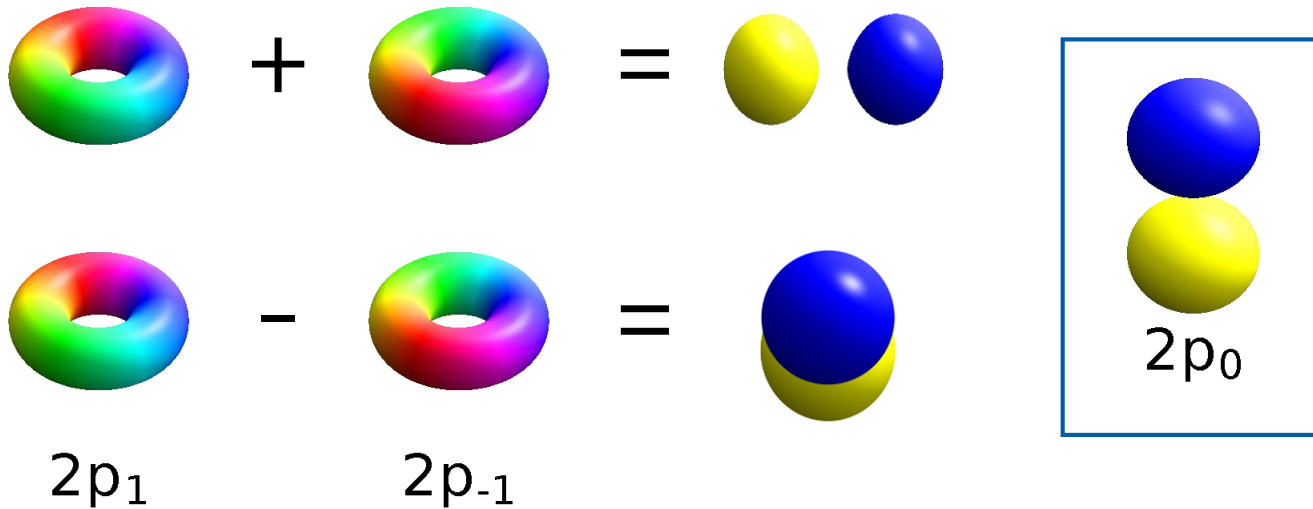
$$\psi_{n11}(r, \theta, \phi) = R_{n1}(r)Y_1^1(\theta, \phi) = -\left(\frac{3}{8\pi}\right)^{1/2} R_{n1}(r) \sin \theta e^{i\phi}$$

$$\psi_{n1-1}(r, \theta, \phi) = R_{n1}(r)Y_1^{-1}(\theta, \phi) = \left(\frac{3}{8\pi}\right)^{1/2} R_{n1}(r) \sin \theta e^{-i\phi}$$

$\psi_{n10}(r, \theta, \phi)$: real / $\psi_{n1\pm 1}(r, \theta, \phi)$: complex

Last class

Mix and Match



(Image: https://commons.wikimedia.org/wiki/File:Atomic_orbitals_n123_m-eigenstates.png)

Still Eigenfunctions?

- np_0, np_1, np_{-1} are all **degenerate** ($E_n = -ZR_H/n^2$)
- Their linear combinations are eigenfunctions of $\hat{H}$ and $\hat{L}^2$, but not those of $\hat{L}_z$

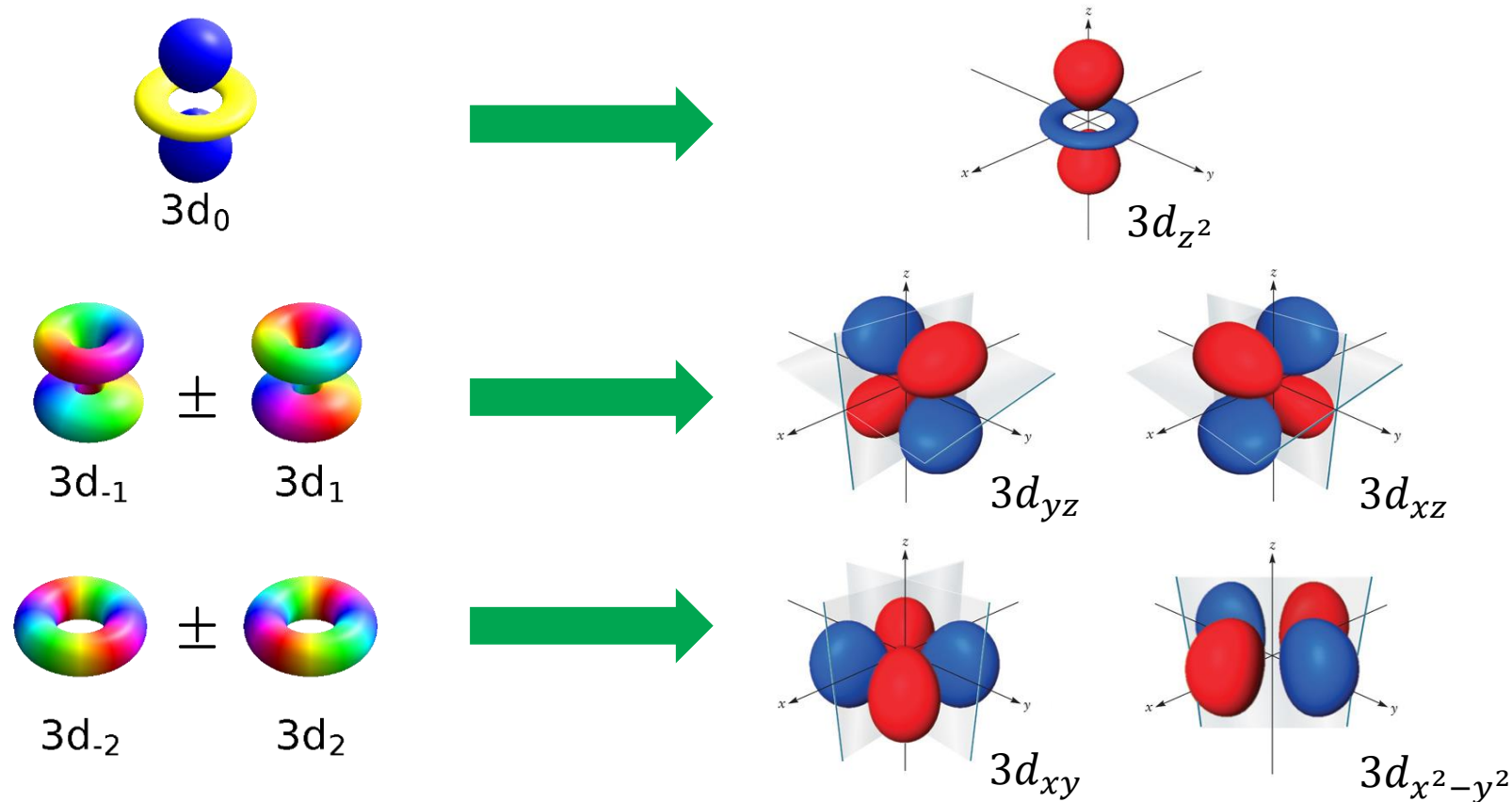
$$\hat{H}\psi_{np_x} = E_n\psi_{np_x}$$

$$\hat{L}^2\psi_{np_x} = l(l+1)\hbar^2\psi_{np_x}$$

$$\hat{L}_z\psi_{np_x} \neq a\psi_{np_x}$$

Last class

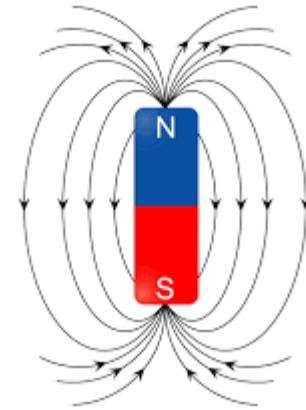
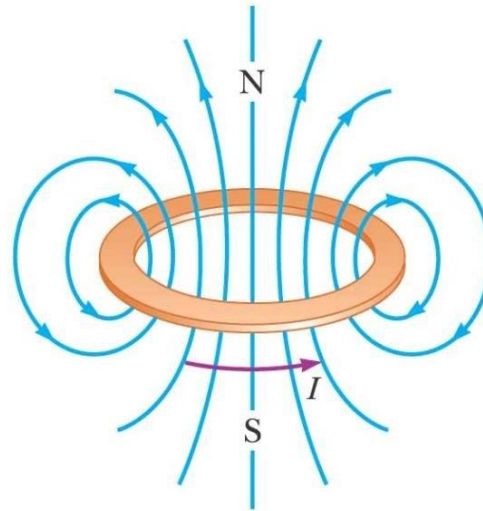
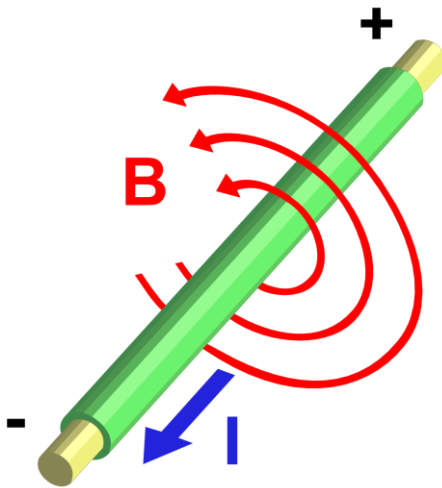
Mix and Match: *d* Orbitals



(Images: https://commons.wikimedia.org/wiki/File:Atomic_orbitals_n123_m-eigenstates.png
Oxtoby et al., *Principles of Modern Chemistry*, 7th Ed., Fig. 5.13)

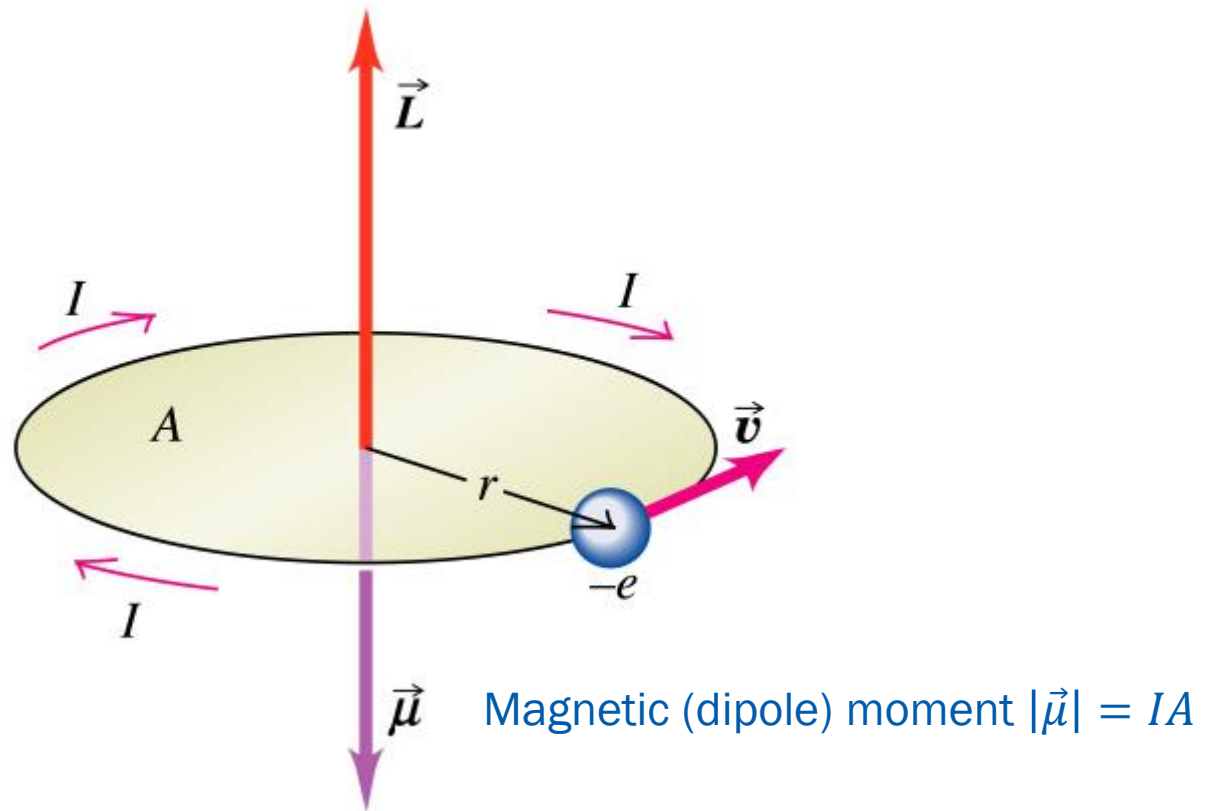
Review

Electric Current & Magnetic Field



(Images: <https://commons.wikimedia.org/wiki/File:Electromagnetism.svg>
<https://www.toppr.com/ask/content/concept/magnetic-field-on-the-axis-of-the-circular-current-carrying-loop-209689/>)

In Case of Electron



(Image: <https://dpotoyan.github.io/Chem324/Lec5-2.html>)

$\vec{\mu}$ and $\vec{L}$

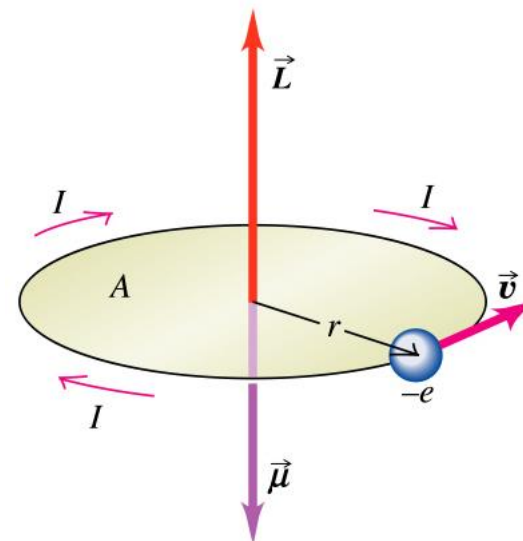
$$I = -\frac{ev}{2\pi r}$$

Combining with $A = \pi r^2$,

$$|\vec{\mu}| = IA = \frac{erv}{2}$$

In general,

$$\vec{\mu} = -\frac{e}{2}(\vec{r} \times \vec{v}) = -\frac{e}{2m}\vec{L}$$



In a Magnetic Field

Potential energy of a magnetic dipole in a magnetic field is

$$V = -\vec{\mu} \cdot \vec{B}$$

Substitution leads to

$$V = -\left(-\frac{e}{2m}\vec{L}\right) \cdot \vec{B} = \frac{e}{2m}\vec{L} \cdot \vec{B}$$

If we set the z-axis as the direction of the external field,

$$V = \frac{eB}{2m}L_z$$

Quantum Mechanics

$$\left(-\frac{\hbar^2}{2\mu} \nabla^2 + \mathcal{V} \right) \psi = E\psi$$

If a hydrogen-atom electron is in a magnetic field,

$$\left(\underbrace{-\frac{\hbar^2}{2\mu} \nabla^2 - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r}}_{\text{Hamiltonian for H atom}} + \underbrace{\frac{eB}{2\mu} \hat{L}_z}_{\text{Additional Hamiltonian}} \right) \psi = E\psi$$

Hamiltonian for H atom
 $\hat{H}_0$

Additional Hamiltonian
 $\hat{H}_{\text{magnet}}$

Do we need to solve the eigenvalue equation again?

Fortunately

The eigenfunctions for $\hat{H}_0$ are also those for $\hat{H}_{\text{magnet}}$

For $\psi_{nlm_l}(r, \theta, \phi) = R_{nl}(r)\Theta_l^{|m_l|}(\theta)e^{im_l\phi}$,

$$\begin{aligned}
 \hat{H}_{\text{magnet}}\psi_{nlm_l} &= \frac{eB}{2\mu}\hat{L}_z\psi_{nlm_l} = -i\hbar\frac{eB}{2\mu}\frac{\partial}{\partial\phi}\psi_{nlm_l} \\
 &= -i\hbar\frac{eB}{2\mu}R_{nl}(r)\Theta_l^{|m_l|}(\theta)\frac{\partial}{\partial\phi}e^{im_l\phi} \\
 &= m_l\hbar\frac{eB}{2\mu}R_{nl}(r)\Theta_l^{|m_l|}(\theta)e^{im_l\phi} \\
 &= \frac{e\hbar}{2\mu}m_lB\psi_{nlm_l} = \beta_B m_l B \psi_{nlm_l}
 \end{aligned}$$

Bohr magneton

New Eigenvalues

$$\left(-\frac{\hbar^2}{2\mu} \nabla^2 - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r} + \frac{eB}{2\mu} \hat{L}_z \right) \psi = E\psi$$

With $\psi = \psi_{nlm_l}$,

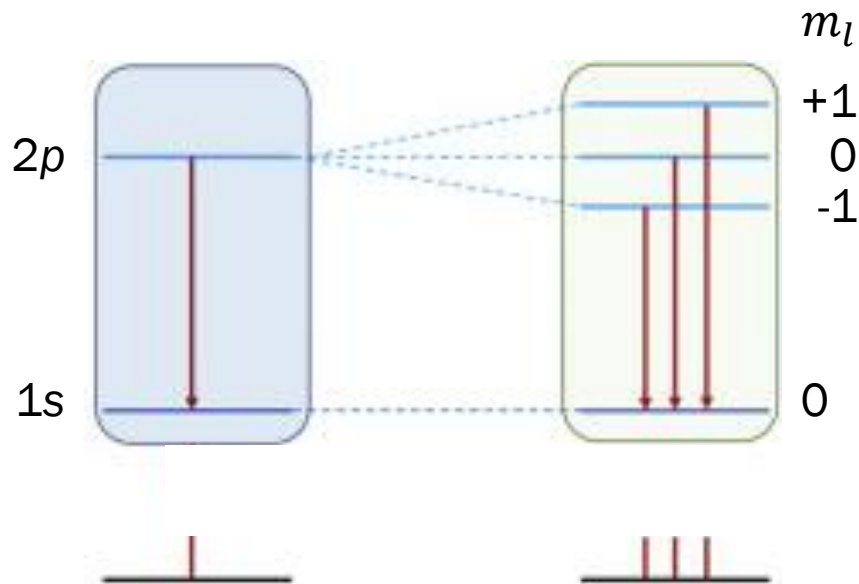
$$E = -\frac{ZR_H}{n^2} + \beta_B m_l B = E_{n,m_l}$$

Now the energy also depends on m_l

(“magnetic quantum number”)

Effect of Magnetic Field

$$E = -\frac{ZR_H}{n^2} + \beta_B m_l B$$

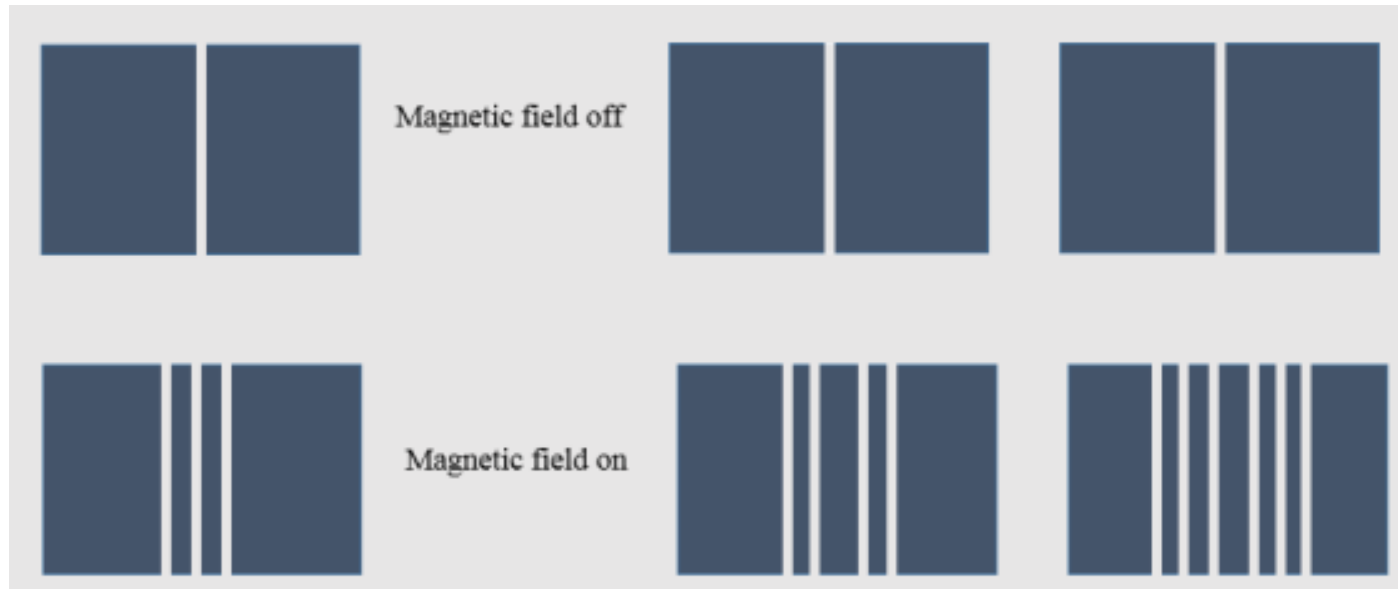


M 7.5

Zeeman Effect

Normal Zeeman effect

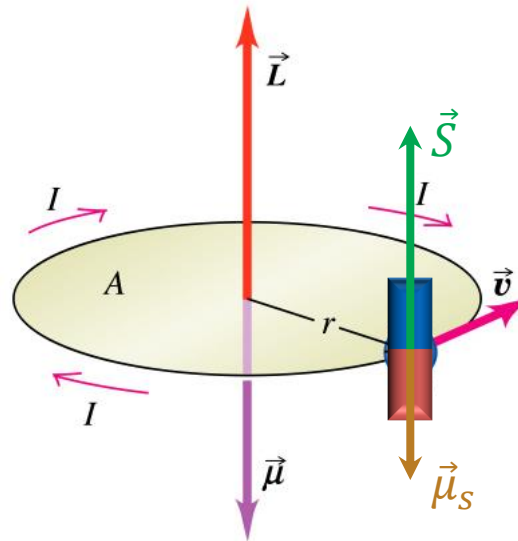
Anomalous Zeeman effect



Intrinsic Angular Momentum

An electron has *intrinsic angular momentum*

= an electron intrinsically behaves like a tiny magnet



(Image: <https://dpotoyan.github.io/Chem324/Lec5-2.html>)

Analogy between Angular Momenta

- Orbital angular momentum: eigenfunction $Y_l^{m_l}$

$$\hat{L}^2 Y_l^{m_l} = l(l+1)\hbar^2 Y_l^{m_l}$$

$$\hat{L}_z Y_l^{m_l} = m_l \hbar Y_l^{m_l}$$

- “Spin” angular momentum: eigenfunction $|s m_s\rangle$

$$\hat{S}^2 |s m_s\rangle = s(s+1)\hbar^2 |s m_s\rangle$$

$$\hat{S}_z |s m_s\rangle = m_s \hbar |s m_s\rangle$$

Analogy between Angular Momenta

- **Orbital angular momentum:** quantum numbers l, m_l

$$l = 0, 1, 2, \dots$$

$$m_l = -l, -(l-1), \dots, (l-1), l$$

- **“Spin” angular momentum:** quantum numbers s, m_s

s : fixed (intrinsic)

$$m_s = -s, -(s-1), \dots, (s-1), s$$

Free Electron in Magnetic Field

- For an electron, $s = 1/2$

$$m_s = \pm 1/2$$

- Orbital angular momentum energy

$$E_{\text{orbital}} = \beta_B m_l B$$

- Spin angular momentum energy

$$E_{\text{spin}} = g\beta_B m_s B$$

- $g \approx 2$ (g-factor): experimentally required

Not in text

Intrinsic Spin Quantum Number s

- Electron: $s = 1/2$
- Proton and neutron: $s = 1/2$
- Photon: $s = 1$
- ^{10}B nucleus: $s = 3$
- ^{12}C nucleus: $s = 0$
- ^{14}N nucleus: $s = 1$
- ^{16}O nucleus: $s = 0$
- ^7Li nucleus: $s = 3/2$
- ^{13}C nucleus: $s = 1/2$
- ^{17}O nucleus: $s = 5/2$
- ^{19}F nucleus: $s = 1/2$

Nuclear Magnetic Resonance

For ^1H and ^{13}C nuclei, $I = 1/2$ and $m_I = \pm 1/2$

(for a nuclear spin, we typically use I instead of s)

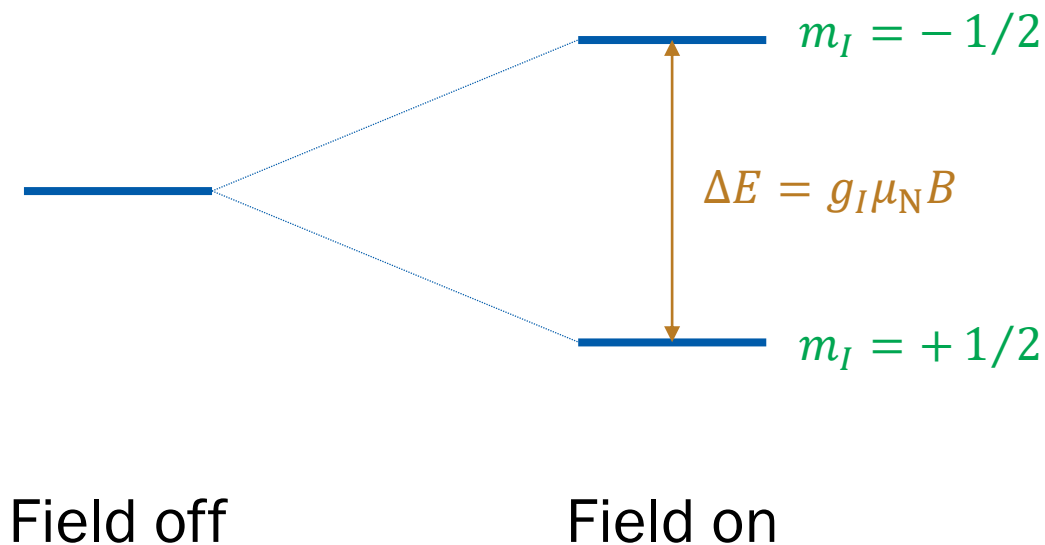
In a magnetic field, the nucleus experiences

$$\hat{H}_{\text{magnet}}|m_I\rangle = -g_I\mu_N m_I B|m_I\rangle$$

where g_I is the nuclear g-factor (experimentally determined)
and μ_N is the nuclear magneton ($e\hbar/2m_{\text{proton}}$)

Nuclear Spin Energy Levels

$$E_{\text{magnet}} = -g_I \mu_N m_I B$$



Not in text

Effect of Chemical Environment

- Electrons can induce an additional magnetic field on top of the applied field
- Local magnetic field: applied field + induced field

$$B_{\text{local}} = B + \delta B$$

- The induced field is proportional to the applied field

$$\delta B = -\sigma B_0$$

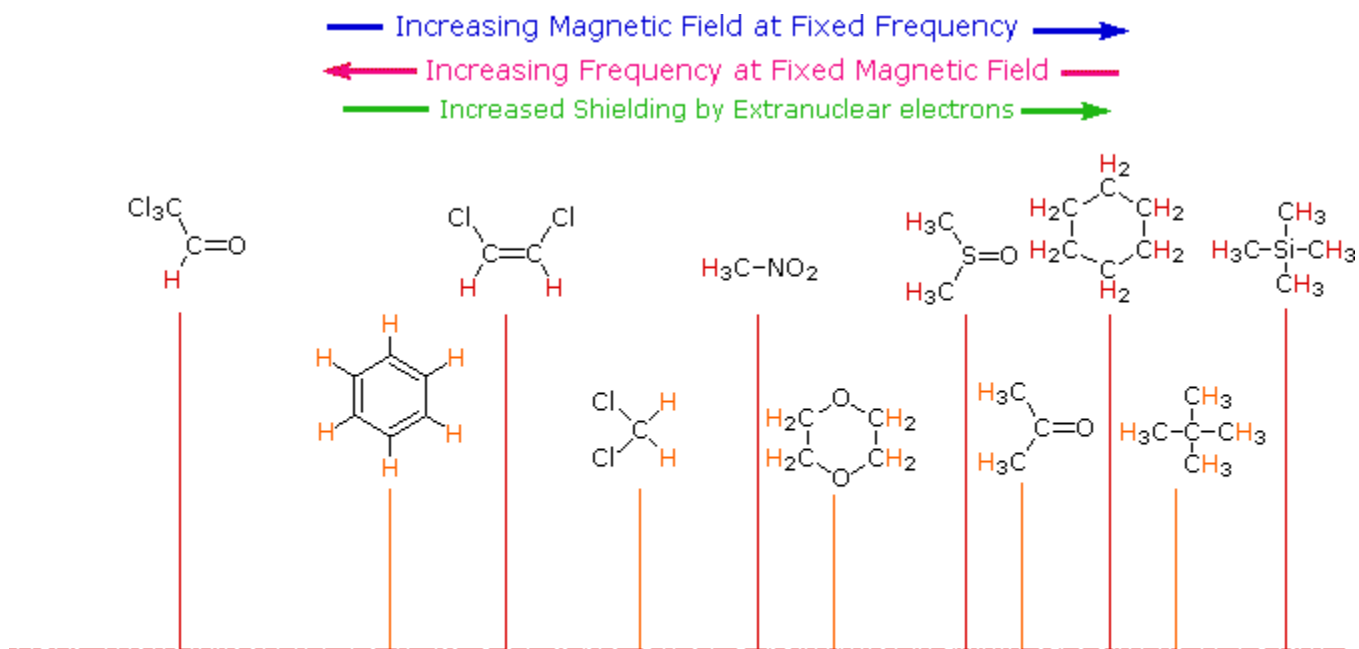
where σ is the shielding constant

- We can measure the shielding constant (“chemical shift”)



Not in text

Chemical Shifts



^1H NMR Resonance Signals for some Different Compounds

(Image: <https://www2.chemistry.msu.edu/faculty/reusch/virttxtjml/spectrpy/nmr/nmr1.htm>)

Not in text

Full Description of H-Atom Electron

- Quantum numbers: n, l, m_l, m_s (s is fixed)
- Wavefunction:

$$\psi_{nlm_l}(r, \theta, \phi) |m_s\rangle = |n \ l \ m_l \ m_s\rangle$$

- Energy (no magnetic field):

$$E_n = -\frac{ZR_H}{n^2}$$

- Energy (in a magnetic field):

$$E = -\frac{ZR_H}{n^2} + ?$$

- Strong field
- Weak field

In a Strong Field

Two magnetic moments independently react to the field

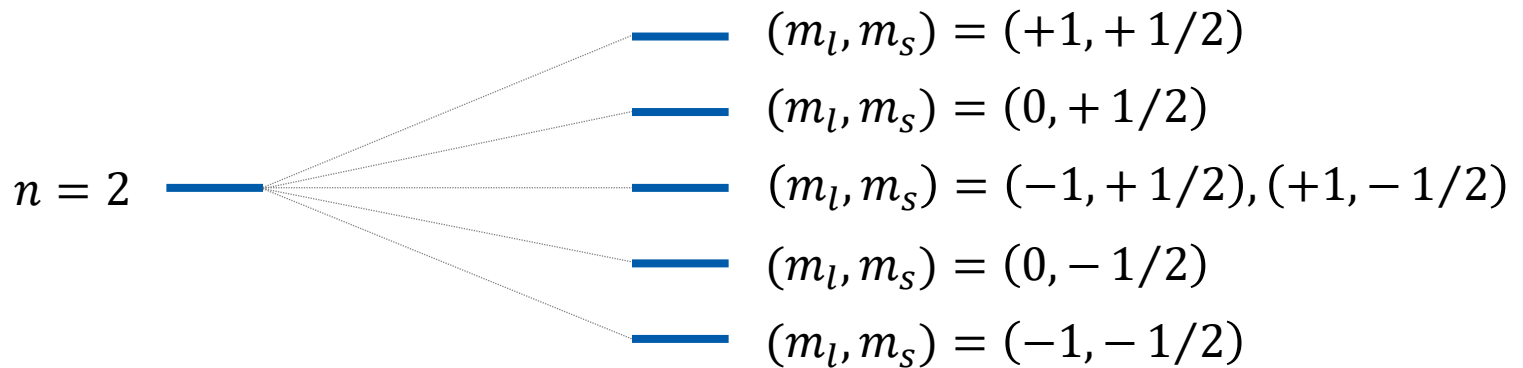
$$E_{\text{magnet}} = E_{\text{orbital}} + E_{\text{spin}}$$

$$E_{\text{orbital}} = \beta_{\text{B}} m_l B$$

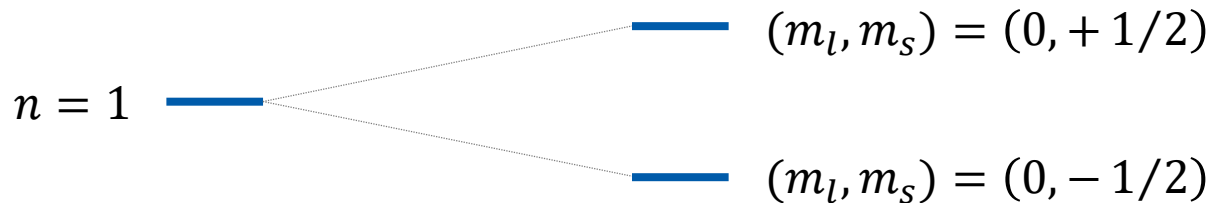
$$E_{\text{spin}} = g \beta_{\text{B}} m_s B$$

$$E = E_n + E_{\text{magnet}} = -Z \frac{R_H}{n^2} + \beta_{\text{B}} B (m_l + 2m_s)$$

Strong-Field Energy Levels



$$E = -Z \frac{R_H}{n^2} + \beta_B B (m_l + 2m_s)$$

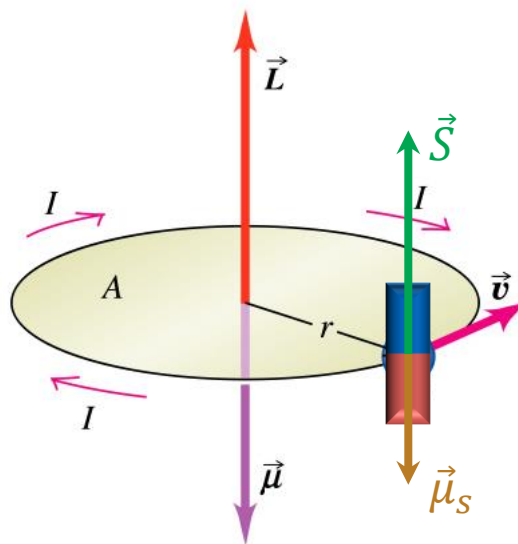


Field off

Field on

In a Weak Field

Need to consider the **spin-orbit interaction**



How?